

Milestone 1

JAVA Lab Resource Utilization Platform

1. Requirement Analysis

Objective

The objective of the Requirement Analysis phase is to understand the project requirements, identify stakeholders, and define the functional and non-functional requirements of the system.

Stakeholders

- Administrator
- Faculty
- Student

Functional Requirements

The system shall provide the following functionalities:

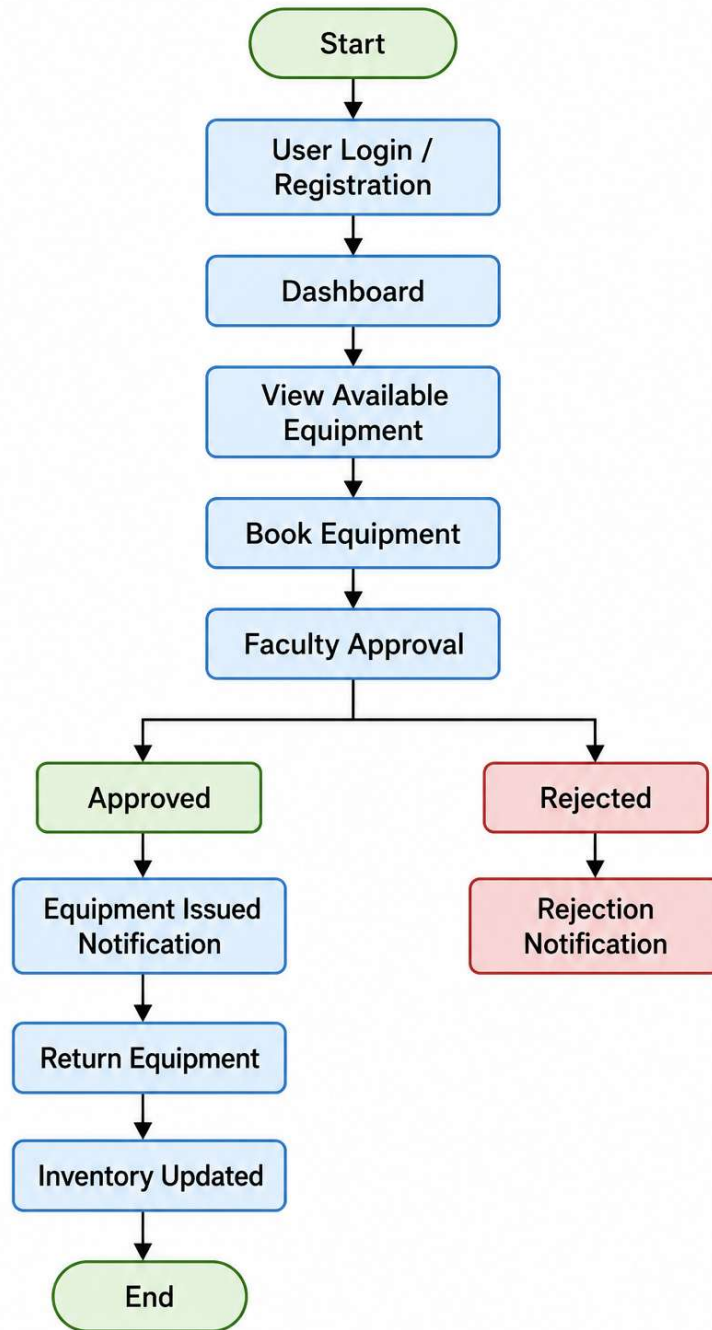
1. User Registration
2. User Login and Authentication
3. Dashboard for different user roles
4. Laboratory Management
5. Equipment Management
6. Equipment Availability Tracking
7. Equipment Booking
8. Booking Approval/Rejection by Faculty
9. Equipment Return Management
10. Booking History
11. User Profile Management
12. Notifications and Alerts

Non-Functional Requirements

1. Secure user authentication
2. Responsive user interface
3. Fast system performance
4. Reliable database management
5. Easy-to-use interface
6. Data consistency and integrity

2. Workflow Design

System Workflow



Workflow Description

Step 1: Login

The user logs into the system using valid credentials.

Step 2: Dashboard

After successful login, the dashboard displays available options according to the user's role.

Step 3: Equipment Availability

Users can browse and search available laboratory equipment.

Step 4: Booking Request

The user submits a booking request by selecting equipment and specifying the required date and time.

Step 5: Faculty Approval

Faculty members review the booking request and either approve or reject it.

Step 6: Equipment Issue

Once approved, the equipment is issued to the student.

Step 7: Equipment Return

After use, the student returns the equipment and the inventory is updated.

3. Documentation

3.1 Project Abstract

The JAVA Lab Resource Utilization Platform is a web-based application developed to simplify the management of laboratory resources in educational institutions. It enables students to book laboratory equipment online, allows faculty members to approve requests, and helps administrators efficiently manage laboratory inventory. The platform reduces manual work, minimizes resource conflicts, and improves the overall utilization of laboratory resources.

3.2 Introduction

Laboratories play an important role in engineering education. Managing laboratory equipment manually often results in misplaced records, scheduling conflicts, and inefficient resource utilization. This project provides an automated solution that digitizes the entire laboratory resource management process, making it more efficient, transparent, and user-friendly.

3.3 Problem Statement

Most educational institutions manage laboratory equipment manually, which causes:

- Time-consuming booking processes
- Duplicate bookings
- Difficulty in tracking equipment availability
- Lack of proper inventory management
- Increased paperwork

3.4 Proposed Solution

The proposed system provides an online platform where students can reserve laboratory equipment, faculty members can approve booking requests, and administrators can manage inventory efficiently through a centralized system.

3.5 Objectives

- Automate laboratory resource management.
- Reduce paperwork.
- Improve equipment utilization.
- Provide real-time equipment availability.
- Simplify booking and approval processes.
- Maintain accurate inventory records.

3.6 Features

- User Authentication
- Role-Based Access
- Laboratory Management
- Equipment Management
- Online Equipment Booking
- Booking Approval System
- Inventory Tracking
- User Dashboard
- Notifications
- Booking History

3.7 Technologies Used

Component	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL
API Testing	Postman
Version Control	Git & GitHub
Design Tool	Figma

3.8 Scope

The system can be implemented in:

- Engineering Colleges
- Universities
- Research Laboratories
- Educational Institutions

3.9 Advantages

- Faster booking process
- Better equipment utilization
- Reduced manual work
- Improved inventory management
- Easy tracking of equipment
- Secure access control
- Paperless management

4. UI Wireframes

The following wireframes have been designed for the project using Figma.

Screen 1 – Login Page

Components:

- College Logo
- Login Title
- Username Field
- Password Field
- Login Button
- Register Link

Screen 2 – Registration Page

Components:

- Full Name
- Email
- Department
- Role Selection

- Password
- Confirm Password
- Register Button

Screen 3 – Dashboard

Components:

- Sidebar Navigation
- Top Navigation Bar
- Statistics Cards
- Recent Bookings
- Notifications

Screen 4 – Equipment List

Components:

- Search Bar
- Equipment Table
- Filter Options
- Equipment Status
- Book Button

Screen 5 – Equipment Booking

Components:

- Equipment Details
- Date Selection
- Time Slot
- Purpose Field
- Submit Booking Button

Screen 6 – Faculty Approval

Components:

- Pending Booking Requests
- Student Details
- Equipment Details

- Approve Button
- Reject Button

Screen 7 – Inventory Management

Components:

- Add Equipment
- Edit Equipment
- Delete Equipment
- Inventory Table

Screen 8 – User Profile

Components:

- Personal Information
- Change Password
- Logout Button

Deliverables

The following deliverables have been completed for Milestone 1:

- Requirement Analysis
- Functional and Non-Functional Requirements
- System Workflow Design
- Project Documentation
- UI Wireframes for all major screens

Conclusion

The completion of Requirement Analysis, Workflow Design, Documentation, and UI Wireframes establishes a strong foundation for the successful development of the JAVA Lab Resource Utilization Platform. These deliverables clearly define the project requirements and provide a structured blueprint for the backend and frontend development phases.